Solutions to selected exercises from §2.4

Question 4

Think about tying and untying your shoes.

Question 7

(a) We prove by constradiction. Suppose there existed A^{-1}. Then from the equation $A^2 = 0$ we can multiply, on the left, by A^{-1} to arrive at the equation $A^{-1}A^2 = A^{-1} \cdot 0$. The right hand side is clearly zero and the left is A, so this implies $A = 0$.

(b) This is impossible. Since A is invertible, we can multiply $AB = 0$ on the left by A^{-1} to arrive at $B = 0$.

Question 10

Define $\mathsf{T}\colon \mathsf{V} \longrightarrow \mathbb{R}^3$ by the formula

$$\mathsf{T}\left(\begin{bmatrix} a & a+b \\ 0 & c \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \tag{1}$$

Note that this map is well-defined! That is, if

$$\begin{bmatrix} a & a+b \\ 0 & c \end{bmatrix} = \begin{bmatrix} a' & a'+b' \\ 0 & c' \end{bmatrix} \tag{2}$$

Then this implies that $a = a', b = b', c = c'$.

Let's first check that this map is injective.

$$\mathsf{T}\left(\begin{bmatrix} a & a+b \\ 0 & c \end{bmatrix}\right) = \mathsf{T}\left(\begin{bmatrix} a' & a'+b' \\ 0 & c' \end{bmatrix}\right) \tag{3}$$

then this means that

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}. \tag{4}$$

So, certainly the original two 2×2 matrices are equal. Thus, the map is injective.

The map is surjective since the dimensions of the domain and codomain agree. So the map is an isomorphism.

Question 16

We first show that Φ is injective. Suppose $\Phi(A) = \Phi(A')$ where A, A' are $n \times n$ matrices. Then $B^{-1}AB = B^{-1}A'B$. Multiplying on the left by B and on the right by B^{-1} we conclude $A = A'$. Again, we apply the dimension theorem to conclude it is an isomorphism.

Question 24

In class on March 16.